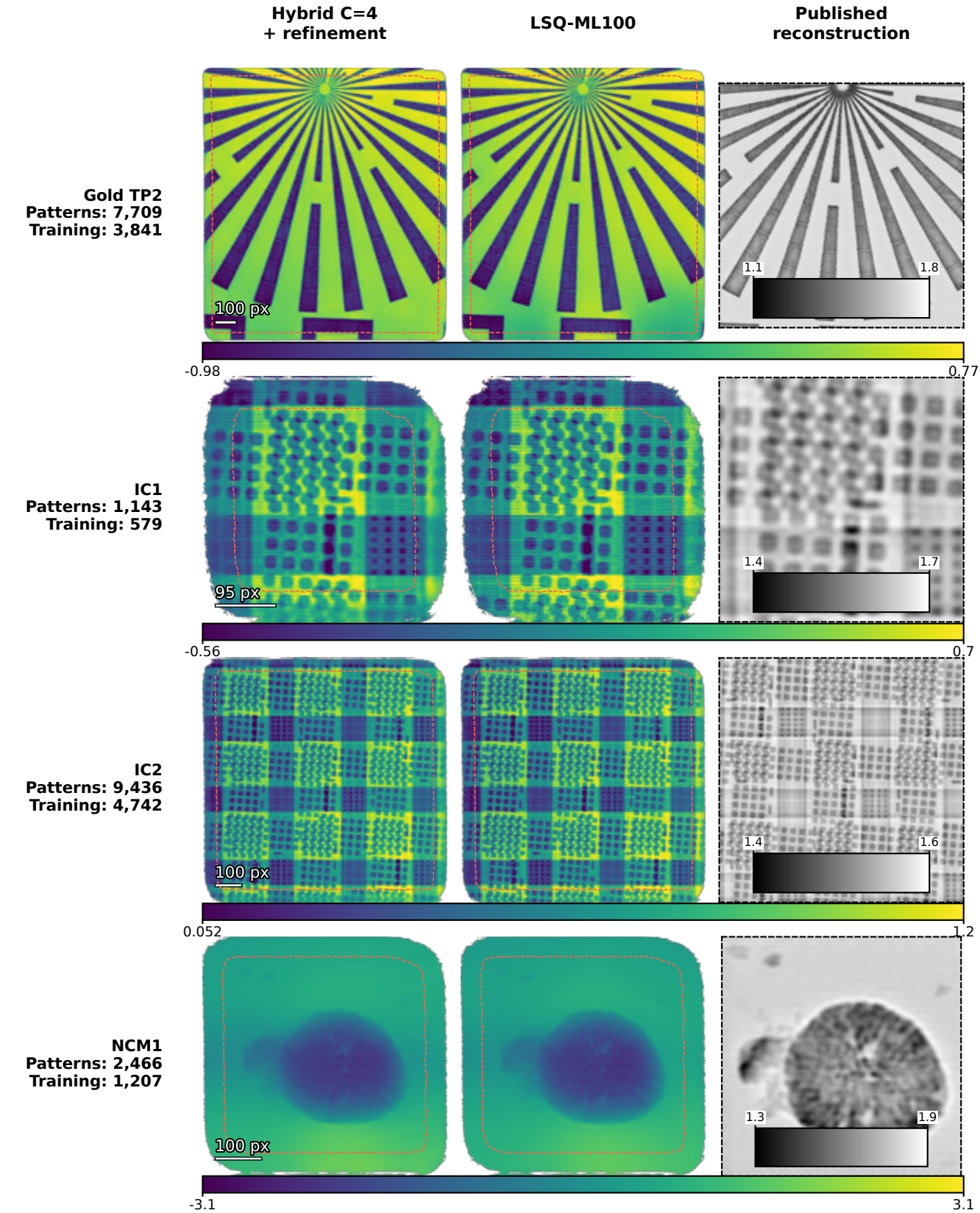


Phase is shown in radians; amplitude shows relative object amplitude. Pattern counts refer to the displayed acquisition and distinct training patterns. High half selects checkpoints (spatial validation, not an independent test; no guard band). Reference is an experimental estimate; dashed contour marks scored interior. Prior fragments: Vong et al., Contrast-invariant deep ptychography neural networks, arXiv:2608.02869, synthetic-only whole-field crops, cropped and rescaled (CC BY 4.0). Their original scales are shown in-cell; the shared computed-method bar excludes prior fragments. Whole-acquisition images include the training region. Paired groups assign C4 only where it has lower phase MAE after refinement and higher phase SSIM; other datasets use C1. This is the C4-selected group. C4 uses 10 groups per training center; training compute is not matched to C1. TP2 is shown full field only; FRC is excluded because of scan-grid artefacts.